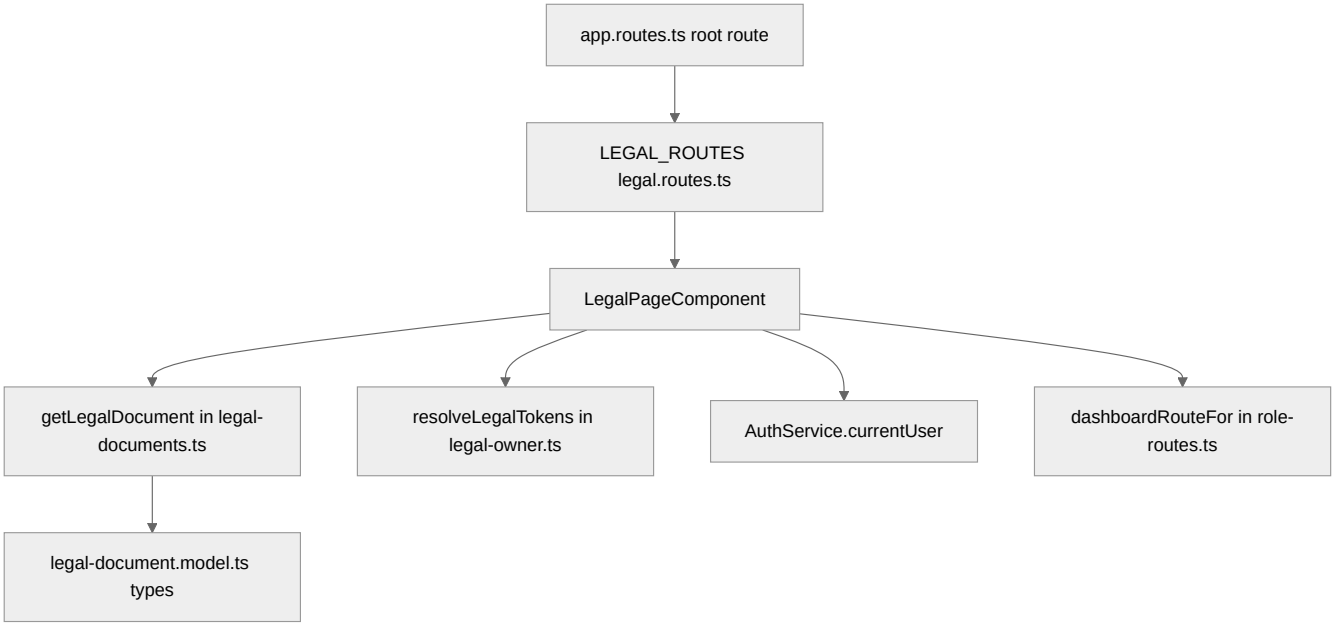
# Legal Documents – Technical Documentation

**Project:** TaskFlow – Intelligent Task Allocation & Tracking System
**Parent Module:** Legal · **Child Module:** 10.1 Legal Documents
**Version:** 1.0 · **Last Updated:** 31-Oct-2026
**Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core)
**Author:** Jojo George (jojogeorge3344@gmail.com)
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## 1. Technical Overview

Legal Documents is frontend-only. Five lazily loaded public routes (`/legal/copyright|terms|privacy|disclaimer|licenses`) all load the standalone `LegalPageComponent` and pass `data.documentId`. The component looks the document up in the `LEGAL_DOCUMENTS` registry (`legal-documents.ts`), replaces the owner tokens with `resolveLegalTokens()` from `legal-owner.ts`, and chooses the back-link target from `AuthService.currentUser()` and `dashboardRouteFor()`. There is no HTTP call, controller, service, repository or table.

## 2. Architecture Flow




## 3. Backend File Structure

| Layer | Path | Files |
|---|---|---|
| Controller / Service / Repository / DTOs / Validators / Entities / Enums / Authorization | *Not applicable for this module.* | – |

No backend file serves the legal documents. Backend files whose behaviour the Privacy Policy text describes are listed in section 9.

## 4. Frontend File Structure

| Type | Path | Purpose |
|---|---|---|
| Routes | `src/app/modules/legal/legal.routes.ts` | `LEGAL_ROUTES`: five routes, each `loadComponent: legalPage` with `data: { documentId }` |
| Root registration | `src/app/routes/app.routes.ts` | `{ path: '', loadChildren: … LEGAL_ROUTES }`, registered **outside** the shell route that has `canActivate: [authGuard, mfaSetupGuard]`, so the pages are public |
| Page component | `src/app/modules/legal/pages/legal-page.component.ts` / `.html` | `LegalPageComponent`: renders one document, the table of contents, the back link and the not-found fallback |
| Models | `src/app/modules/legal/models/legal-document.model.ts` | `LegalDocumentId`, `LegalBlock` (`paragraph` / `list` / `note` / `table`), `LegalSection`, `LegalDocument` |
| Data / utils | `src/app/modules/legal/utils/legal-documents.ts` | `LEGAL_DOCUMENTS` (660-line registry), `getLegalDocument(id)`, `LEGAL_DOCUMENT_IDS` |
| Data / utils (shared with 10.2) | `src/app/modules/legal/utils/legal-owner.ts` | `LEGAL_OWNER`, `LEGAL_APPLICATION`, `LEGAL_TOKENS`, `resolveLegalTokens(text)`, `legalOwnershipLine()` |
| Helpers used | `src/app/shared/auth/auth.service.ts`, `src/app/shared/auth/role-routes.ts` | `AuthService.currentUser()`, `dashboardRouteFor(role, permissions = [])` |
| Route / Guard | *None* | No guard and no permission check. No entry in `layouts/nav-items.ts` |

## 5. API & Logic

### 5.1 API Endpoints

*Not applicable for this module.* (No backend endpoints. The content is bundled in the frontend.)

Client routes (Angular, from `legal.routes.ts`):

| # | Route | `data.documentId` | Document title | Footer label | Sections |
|---|---|---|---|---|---|
| 1 | `/legal/copyright` | `copyright` | Copyright Notice | Copyright | 8 |
| 2 | `/legal/terms` | `terms` | Terms of Use | Terms of Use | 14 |
| 3 | `/legal/privacy` | `privacy` | Privacy Policy | Privacy Policy | 11 |
| 4 | `/legal/disclaimer` | `disclaimer` | Disclaimer | Disclaimer | 7 |
| 5 | `/legal/licenses` | `licenses` | Third-Party Licence Notices | Licences | 3 |

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All five documents have `updated: '21 October 2026'`.

## 5.2 Request / Response Samples

*Not applicable for this module.* The data shape is the TypeScript model below (extract from a registry entry):

```
{
  "id": "copyright",
  "title": "Copyright Notice",
  "navLabel": "Copyright",
  "summary": "Who owns {{applicationName}}, what this notice protects and how you may use it.",
  "updated": "21 October 2026",
  "sections": [
    { "id": "ownership", "heading": "Ownership",
      "blocks": [ { "kind": "paragraph", "text": "The owner can be contacted at {{email}}." } ]
}
  ]
}
```

## 5.3 Service Layer Logic

*Not applicable for this module.* (No backend service.) The frontend logic in `LegalPageComponent` is:

1. `routeData` is `toSignal(route.data.pipe(map(d => d['documentId'])))`.
2. `document` is `computed`. It calls `getLegalDocument(id)` (`LEGAL_DOCUMENTS.find(doc => doc.id === id)`) and returns `undefined` when there is no id or no match.
3. `resolvedDocument` is `computed`. It deep-clones the document (`JSON.parse(JSON.stringify(doc))`) and runs `resolveLegalTokens()` over `title`, `summary`, `navLabel`, every section `heading`, paragraph/note `text`, list `items`, and table `caption`, `head` and `rows`.
4. `resolveLegalTokens(text)` loops over the `TOKEN_VALUES` map and calls `replaceAll(token, value)`.
5. `ownershipLine` comes from `legalOwnershipLine()`, which returns `` `© ${year} ${name}. ${rights}.` ``, giving "© 2026 Jojo George. All Rights Reserved.".
6. `backLink` is `computed`. If `auth.currentUser()` exists it returns `{ path: dashboardRouteFor(user.role), label: 'Back to the application' }`, otherwise `{ path: '/login', label: 'Back to sign in' }`. `dashboardRouteFor` is called **without** the permissions argument, so the result is Administrator → `/admin`, Manager → `/manager`, any other role → `/me`.
7. The template renders `@if (resolvedDocument(); as doc)` → header, `nav.legal__toc` (anchor `href="#" + section.id`), `article` sections by block kind (`<p>`, `<aside class="legal__note">`, `<ul>`, `<table>` with `<caption>`/`<th>`). `@else` renders the "Legal document not found" fallback.
8. Track functions: `trackSection` → `section.id`, `trackBlock` → `block.kind`, `trackItem` → index.

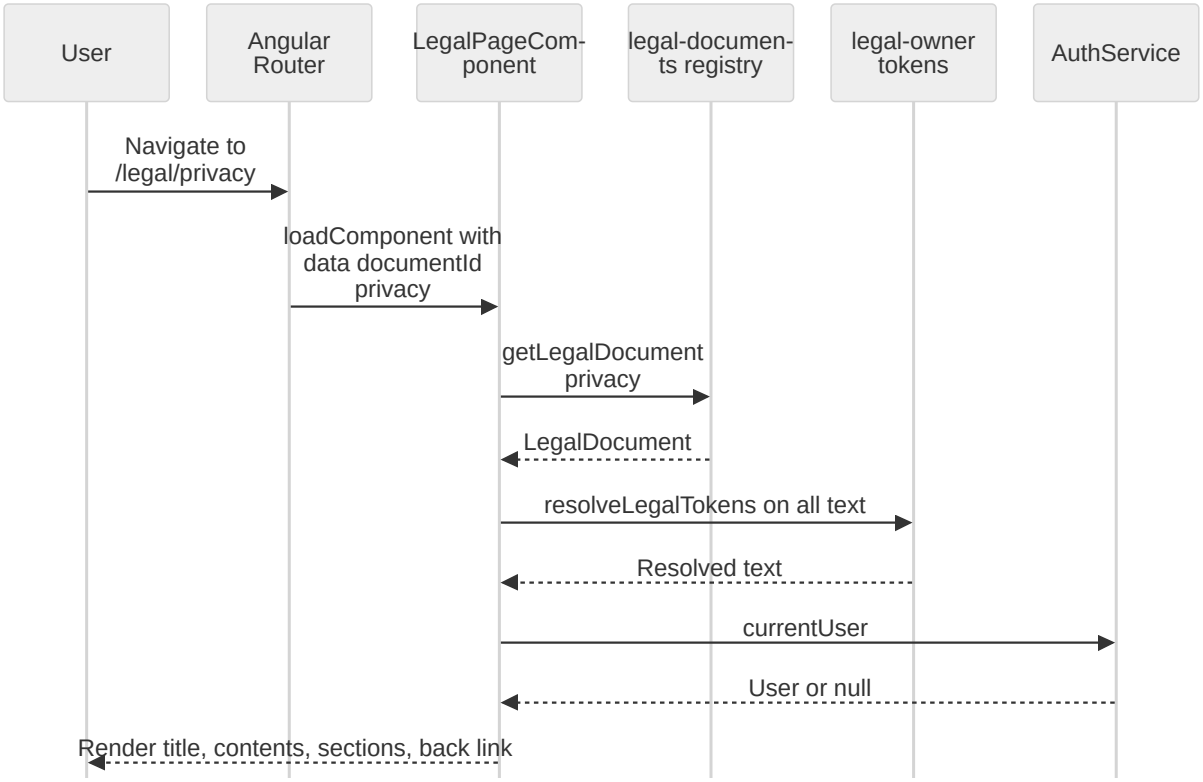
## 5.4 Repository Layer

*Not applicable for this module.*

| Method | Description | Query Notes |
|---|---|---|
| `getLegalDocument(id)` (frontend util, not a repository) | Looks up a document by id in `LEGAL_DOCUMENTS` | In-memory `Array.find` |

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### 5.5 Sequence Diagram




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## 6. Database Design

*Not applicable for this module.* (No tables, entities or migrations. The five documents are static data in `legal-documents.ts`.)

**Relationships**

- *Not applicable for this module.*

## 7. Security & Authorization

- **Authentication:** Not required. `LEGAL_ROUTES` is registered at root level, outside the shell route that has `authGuard` and `mfaSetupGuard`. `AuthService.currentUser()` is read only to choose the back-link target.
- **Authorization:** No permission key for the legal pages.
- **Output safety:** All document text is rendered with Angular interpolation (`{{ }}`), so it is HTML-escaped. `innerHTML` is not used.

| Permission Key | Endpoint(s) | UI Element Controlled |
|---|---|---|
| *(none)* | – (client routes `/legal/copyright`, `/legal/terms`, `/legal/privacy`, `/legal/disclaimer`, `/legal/licenses`) | Legal document pages are always visible (Always On for every role, and also for visitors who are not signed in) |

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## 8. Validations & Error Handling

| Rule | Layer | HTTP Code | Error Message |
|---|---|---|---|
| Route has no `documentId` or the id is not in the registry | `LegalPageComponent` (`document()` returns `undefined`) | – (client) | "Legal document not found" / "The document you requested is not available." |
| Unknown `/legal/xyz` path | `app.routes.ts` wildcard `{ path: '**', redirectTo: 'login' }` | – (client) | Redirects to `/login` (no message) |

## 9. Integrations / Side Effects

- **Notifications / Email / Activity Log:** *Not applicable for this module.* Reading a document saves nothing.
- **Backend facts the Privacy Policy states, checked against the code:**
  - Default storage root is `C:\TaskFlow` on Windows, otherwise `CommonApplicationData/TaskFlow` (`Persistence/StorageRoot.cs`). Program.cs references `taskflow.db`, `dp-keys` and `mail-attachments`.
  - `IPasswordHasher<User>` is registered as `PasswordHasher<User>` (`Config/IntegrationRegistration.cs`).
  - Data Protection uses `ProtectKeysWithDpapi()` (`Config/IntegrationRegistration.cs`).
  - The cloud-model notice exists in the assistant panel: `assistant.needsCloudNotice()` in `modules/assistant/components/tfa-panel.component.html`.

## 10. Configuration

| Key (`appsettings.json`) | Purpose | Default |
|---|---|---|
| `Storage:Root` (referenced by the Privacy Policy text only) | Data folder. Empty means the platform default | `""` |
| `AiProviders:KeyRingPath` (referenced by the Privacy Policy text only) | Data Protection key ring folder | `"dp-keys"` |

Owner values come from the compile-time constants in `legal-owner.ts` (see the parent document, section 10).

## 11. Testing

| Test Type | File / Location | Covers |
|---|---|---|
| Unit | *None found* | `tests-filelist.json` (backend tests) has no Legal test. No frontend spec files were supplied. |
| Integration | *None found* | – |

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> ⚠️ To be confirmed: whether a frontend spec such as `legal-page.component.spec.ts` exists, because frontend test files are not in the supplied file list.

## 12. Known Limitations / Gaps found

- The content is hard-coded in `legal-documents.ts`. Any wording change needs a frontend rebuild, and there is no admin editor or versioning. The "Last updated" date (`21 October 2026`) is typed by hand for each document.
- There is no acceptance tracking. The Terms say "By opening, installing or using the application you accept these terms", but no consent is recorded.
- The third-party licence table is maintained by hand and has to be kept in step with the real `package.json` and `.csproj` files.
- Possible enhancement: serve the documents from the backend (or a settings table) so they can be updated without a rebuild.

**Gaps found**

- **Back link for custom roles (confirmed, matches `permission-gaps.md`):** `backLink` calls `dashboardRouteFor(user.role)` without permissions, so a custom role that holds `dashboard.view.organization` or `dashboard.view.team` is sent to `/me` instead of `/admin` or `/manager`.
- **Not-found fallback cannot be reached by URL:** only the five known ids have routes. Any other `/legal/...` path matches the `**` wildcard and redirects to `/login`, so the "Legal document not found" branch runs only if a route is added without a matching registry entry.
- **Duplicate track keys:** `trackBlock` returns `block.kind`. Sections with two blocks of the same kind (for example Copyright → Ownership has two paragraphs) produce duplicate `@for` track keys, and Angular warns about duplicate keys in dev mode.
- **Missing stylesheet file:** `styleUrl: './legal-page.component.css'` is referenced but not in the supplied tree (module-wide note in the parent document).
- **Referenced file not in tree:** the Licences page says the full texts are in `THIRD-PARTY-NOTICES.md` at the repository root. No such file is in the supplied tree.

  > ⚠️ To be confirmed.

- **Licence table versions not verifiable:** the table lists Angular 19, TypeScript 5.7.x, RxJS 7.x, Zone.js 0.15.x and .NET / ASP.NET Core / EF Core 10. No `package.json` or `.csproj` was supplied to check them.

  > ⚠️ To be confirmed.

- **Content wording issues:**
  - The Terms "Disclaimer of warranties" paragraph repeats "fitness for a particular purpose" twice.
  - The Terms "Changes to these terms" section says the date is "at the foot of this document", but the page shows "Last updated" in the document header.
  - The Licences page says the TFA name and TF mark "are used under permission", which reads oddly given that the owner holds them.

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